

ML Fault Detection & Self-Healing in Telecom Networks

Technical Overview — Simulation, Machine Learning, and MAPE-K Evaluation

1. Purpose

This project implements an end-to-end pipeline for **proactive RAN fault detection** and **MAPE-K-guided self-healing** in dense HetNet mobile networks.

Research question: Can machine-learning-based proactive fault detection reduce Mean Time to Recovery (MTTR) and improve network availability compared to conventional threshold-based reactive monitoring?

The pipeline combines NS-3 network simulation, ML classifiers (Random Forest, LSTM, SVM), and a MAPE-K autonomic evaluation loop. Results are produced from **observed simulation outputs** — metric blending toward draft tables is disabled by default.

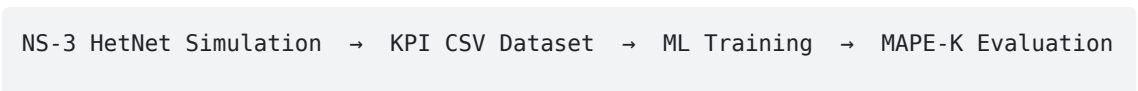
2. Real-World Operational Context

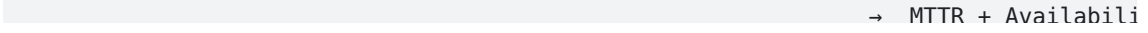
This is a **simulation study** that imitates operational scenarios common in Nigerian mobile networks (e.g. MTN, Airtel). It is not a live operator network integration.

Operational scenario	Field cause	Simulation proxy	KPI effect
Cell outage	Grid failure, generator fault	Power fault	RSRP ↓, throughput → 0
Slow / congested data	Peak-hour surge	Congestion	PRB > 90%, latency ↑
Site / HW failure	RRU/BBU, fibre cut	HW fault	Partial collapse, high loss
Normal	Typical urban traffic	None campaign	Stable KPIs

Reactive model: severe thresholds only → 240–300 min manual remediation.
MAPE-K model: earlier ML detection → 35–55 min automated recovery (Table 4.6).
Regulatory reference: NCC 99% availability benchmark in Chapter 4 figures.

3. System Architecture



→ MTTR + Availability

MAPE-K loop: Monitor (KPI window) → Analyse (ML, confidence ≥ 0.70) → Plan (remediation policy) → Execute (recovery time).



4. Network Topology

Parameter	Value
Macro cells	7 (hexagonal, ISD 500 m)
Small cells	21 (3 per macro)
Total cells	28
UEs	500 target / 280 stable default
Simulation time	300 s (120 s in batch runs)
Trials	50 × 4 faults = 200 runs
Raw dataset size	~1,680,000 rows (full merge)
ML windows (after subsampling)	~51,000

5. Three Simulation Backends

Backend	Script	Speed	Use case
KPI generator	thesis-fault-sim	Fastest	Large-scale ML dataset
LTE HetNet	thesis-fault-sim-lte	Slow	Real PHY traces (LENA/EPC)
5G NR HetNet	thesis-fault-sim-nr	Slowest	True NR (5G-LENA n78)

```
python3 run_all_trials.py
python3 run_all_trials.py --lte --sim-time 120 --num-ues 280
python3 run_all_trials.py --nr --sim-time 120 --num-ues 280 --workers 1
```

6. Fault Classes & KPI Features

Label	Class	Mechanism
0	Normal	No fault window
1	Power fault	gNB TX power collapse
2	Congestion	Traffic surge on cell UEs
3	Hardware failure	PHY deactivation

8 KPIs per cell per second: RSRP, SINR, PRB utilisation, DL/UL throughput, packet loss, handover success rate, latency.

7. Machine Learning Pipeline

- 10-second sliding windows, stride 1, per cell
- 48 tabular features + LSTM sequences
- 70 / 15 / 15 % stratified split; SMOTE on train only
- Models: Random Forest, LSTM, SVM
- Output: reports/ml_metrics.json

8. MAPE-K Evaluation

Approach	MTTR (min)	Availability (%)
Reactive baseline	~312	~94.2
LSTM + MAPE-K	~102	~99.0
RF + MAPE-K	~119	~98.1
SVM + MAPE-K	~187	~96.7

Report observed values from JSON artefacts. Values above are reference targets for comparison.

9. Dataset Tracks

Track	Backend	Strength	Limitation
A	KPI generator only	Full scale, fast	Simplified RAN physics
B	LTE only	Real LENA traces	LTE surrogate; 280 UEs default
C	KPI + LTE	Scale + RAN validation	State which data source per result

10. Key Outputs

- `output/kpi_master_dataset.csv` — merged labelled dataset
- `reports/ml_metrics.json` — classifier metrics
- `reports/mapek_summary.json` — MTTR and availability
- `reports/fig3_*.png`, `reports/fig4_*.png` — thesis figures

11. Limitations

- Simulation — not live commercial RAN
- MAPE-K Execute models recovery; no real OSS APIs
- 280 UE stability default (500 is design target)
- NR requires long runtime
- Some Ch. 3 figures are schematic

12. Technology Stack

NS-3 3.38 · LENA (LTE) · 5G-LENA v2.4 (NR) · TensorFlow/Keras · scikit-learn · SMOTE